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H2

Started on	Monday, 17 August 2026, 10:38 PM
State	Finished
Completed on	Monday, 17 August 2026, 10:42 PM
Time taken	4 mins 46 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c = 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2  int main()
3  {
4      int n,c=0;
5      scanf("%d",&n);
6      c+=2;
7      for(int i=n/2;i<n;i++)
8      {
9          c+=2;
10         for(int j=1;j<n;j=j*2)
11         {
12             c+=2;
13             for(int k=1;k<n;k=k*2)
14             {
15                 c+=2;
16             }
17         }
18     }
19     printf("%d",c);
20     return 0;
21 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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